

Making Charts Speak: LLM-Based Conversational Chart Question Answering for Blind and Low-Vision Users

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Abstract

Blind and low-vision (BLV) users face significant barriers when encountering charts in digital content. While recent advances in large language models (LLMs) show promise for chart interpretation, existing accessible systems rely on pre-structured data formats, limiting applicability to real-world charts. We present GraphWhisper, a conversational system enabling BLV users to explore charts through natural language without requiring pre-structured data. GraphWhisper employs enhanced chart-specific prompting strategies that guide LLMs to systematically extract and analyze data directly from chart images, achieving 98.1% accuracy on visualization literacy benchmarks. A user study with 15 BLV participants demonstrated 94% question-answering accuracy across 185 questions and exceptional satisfaction scores (4.87-5.0 on 5-point scales), validating the system's practical effectiveness for everyday chart accessibility.

CCS Concepts

• **Human-centered computing** → **Accessibility; Empirical studies in accessibility; Visualization systems and tools; • Computing methodologies** → **Natural language processing.**

Keywords

Chart Accessibility, Blind and Low-Vision Users, Large Language Models, Data Visualization

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1 Introduction

Data visualizations are fundamental to information communication across digital platforms, yet they create significant accessibility barriers for the estimated 285 million people worldwide who are blind or have low vision (BLV) [33]. Current accessibility standards provide limited solutions: alternative text descriptions reduce rich

visual information to static summaries, while data tables require manual navigation without visual context [1, 23, 34].

Recent advances in large language models (LLM) [7, 8] have opened new possibilities for chart accessibility [22, 36]. Kim et al. [17] established through a Wizard of Oz study with 24 BLV participants that value retrieval and extrema finding are the most common analytical tasks. Building on these insights, VizAbility introduced an LLM-based conversational system [12], achieving 87.39% query classification accuracy. However, VizAbility requires chart data in structured formats (Vega-Lite specifications), limiting applicability to real-world scenarios where charts exist as image files.

Despite these advances, a significant gap remains between research prototypes and practical accessibility solutions. Existing systems either require pre-structured data formats that limit real-world applicability [12] or focus primarily on evaluation benchmarks rather than user-centered design [9]. BLV users encounter charts daily in formats that current systems cannot process: PNG images in news articles, JPEG figures in research papers, and embedded graphics in web content. Furthermore, current systems lack the contextual support features that BLV users need for effective chart exploration, including automatic summaries, suggested follow-up questions, and confidence indicators that help assess system reliability [17].

To address these limitations, we present GraphWhisper, a conversational chart accessibility system that processes standard image formats (JPEG, PNG) while maintaining high accuracy. We pose the following research questions:

- **RQ1:** Can enhanced prompting strategies that incorporate chart-type-specific analysis improve LLM performance on chart interpretation tasks beyond current state-of-the-art approaches?
- **RQ2:** How can we design a practical chart accessibility system that processes standard image formats while achieving question-answering accuracy comparable to structured-data systems, and what are the practical trade-offs of this approach?
- **RQ3:** What interface features and interaction patterns support BLV users in exploring charts through conversational interaction with image-based content?

Our key contributions are: (1) Enhanced chart-type-specific prompting methodology achieving 98.1% accuracy on VLAT benchmarks, surpassing previous state-of-the-art results; (2) A practical conversational interface with automatic summarization, dynamic follow-up questions, and confidence indicators; (3) A user study with 15 BLV participants demonstrating 94% question-answering



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accuracy and exceptional satisfaction (4.87-5.0/5.0 across usability dimensions).

2 Related Work

Traditional chart accessibility approaches focus on alternative text descriptions and data tables [27], which fail to capture the exploratory benefits of visualizations [23]. Alternative text provides only high-level descriptions, while data tables require manual navigation without visual context [1]. Researchers have explored non-visual modalities including sonification systems [11, 35], which struggle with complex visualizations and require specialized training [29], and tactile approaches [4, 6], which need specialized hardware and fail to scale to real-world complexity [5]. Keyboard-navigable representations like Olli [2], Data Navigator [10], and MAIDR [30] offer more practical alternatives but present challenges for efficient data exploration [18], with users struggling with orientation and pattern extraction from sequential navigation [14].

Chart question answering systems have evolved from template-based approaches [15, 16] to sophisticated models incorporating real-world charts [25, 26]. However, most research focuses on general populations rather than BLV users' specific needs. Kim et al. [17] conducted the first systematic study with 24 BLV participants, collecting 979 queries and establishing that value retrieval and extrema finding are the most common tasks. VizAbility [12] introduced the first working LLM-based conversational system, achieving 87.39% query classification accuracy but requiring structured Vega-Lite inputs, limiting applicability to real-world image-based charts.

The Charts-of-Thought method [9] introduced systematic prompting for chart interpretation, achieving 50.17 on VLAT with Claude-3.7, surpassing the human baseline of 28.82. However, it employs uniform data extraction procedures across all chart types, not accounting for unique visual challenges—pie charts require geometric estimation, stacked charts need boundary calculations, and choropleth maps demand color discrimination strategies [19, 21]. Our work builds on this foundation by developing chart-type-specific prompting strategies that adapt the analysis process to unique characteristics of different visualization types.

3 Enhanced Charts-of-Thought Methodology

We developed an enhanced prompting strategy building on Charts-of-Thought [9], introducing chart-type-specific data extraction procedures and enhanced verification. While the original method employs uniform procedures across all chart types, we recognized that different visualizations present unique analytical challenges requiring specialized approaches.

3.1 Chart-Type-Specific Analysis

Our methodology begins with explicit chart type identification, prompting the LLM to classify the uploaded image before proceeding with analysis. We expanded the taxonomy to twelve types: Line Chart, Bar Chart, Stacked Bar Chart, 100% Stacked Bar Chart, Pie Chart, Histogram, Scatterplot, Bubble Chart, Area Chart, Stacked Area Chart, Choropleth Map, and Treemap.

We developed specialized data extraction procedures addressing each chart type's unique characteristics. For stacked charts,

our procedures include explicit instructions for calculating values using boundary differences (Top Boundary - Bottom Boundary), preventing errors where LLMs incorrectly use boundary values as data values. For pie charts without explicit values, we introduced geometric degree estimation with mathematical verification: ranking slices by size, estimating degrees, and verifying totals sum to 360 degrees. Our choropleth map procedures address complex visual interpretation challenges, including handling tied regions with explicit tie-breaking rules.

3.2 Validation and Verification

Our methodology incorporates validation guardrails within the prompting structure to ensure correctness. For percentage-based charts (pie charts, 100% stacked charts), the LLM verifies that extracted values sum to the expected total (100% or 360 degrees). For stacked visualizations, the system validates that segment calculations follow the formula (Top Boundary - Bottom Boundary) and cross-checks that individual segment values sum to the total bar height. These programmatic validation steps are embedded as explicit verification instructions in our chart-type-specific prompts, enabling the LLM to self-correct extraction errors before generating final responses.

3.3 Evaluation Results

We evaluated using the original VLAT benchmark [20] and a modified version [9], testing across Claude 4.5, GPT-5, and Gemini 2.5. Claude 4.5 achieved 52/53 questions correct on both datasets (98.1% accuracy), surpassing the original Charts-of-Thought (50.17) and human baselines (28.82). Our enhanced method also improved GPT-5 (from 46.00 to 49.00 on original, 45.00 to 47.00 on modified) and Gemini 2.5 (from 41.00 to 45.00 on original, 46.67 to 47.00 on modified). Additional validation on ChartQA [25], ChartMuseum [31], and EncQA [28] benchmarks confirmed generalization. Complete results appear in the appendix.

4 System Design

GraphWhisper employs a comprehensive processing pipeline transforming uploaded chart images into accessible conversational interfaces (Fig. 1).

4.1 Design Goals

GraphWhisper addresses three primary goals: (1) Support real-world formats (JPEG, PNG); (2) Comprehensive query support [17]; (3) Enable independent exploration with automatic summaries and confidence indicators.

4.2 System Architecture

The system operates through four main stages (Fig. 1):

(1) Image processing: The system validates input formats and applies enhanced Charts-of-Thought methodology for chart interpretation. Chart type identification occurs first, determining which specialized analysis procedures to apply. The system extracts visual elements, identifies data encodings, and interprets axis labels and legends.

(2) Analysis: The system generates automatic summaries identifying chart type, describing data ranges, highlighting key patterns,

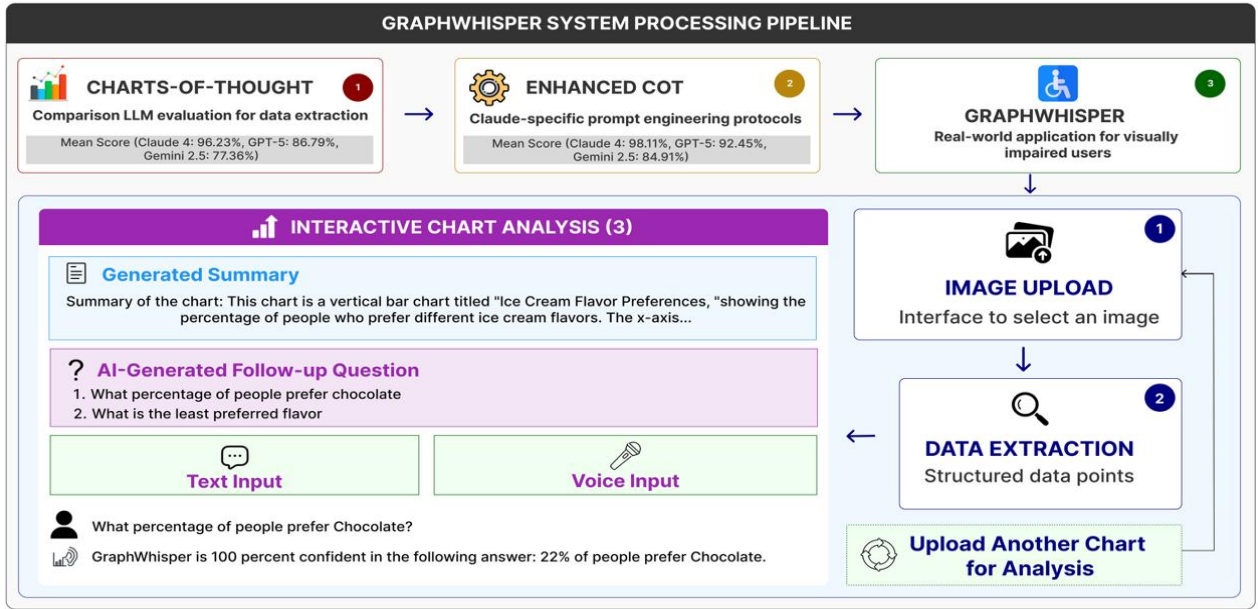


Figure 1: GraphWhisper system processing pipeline showing the complete workflow from image upload to user interaction.

and explaining visual encodings. For charts with multiple data series, summaries explain relationships between datasets and visual encoding differences. Summary detail adapts to chart complexity.

(3) **Query processing**: Incoming questions are classified using Kim et al.'s taxonomy [17] through few-shot prompting with examples from their corpus of 979 real BLV user questions. Analytical queries trigger data extraction procedures, visual queries activate descriptive capabilities, contextual queries integrate external knowledge, and navigation queries provide exploration guidance.

(4) **Response generation**: The system provides answers with confidence indicators through multimodal interface options, assessing certainty based on data extraction clarity, question complexity, and image quality.

4.3 Core Interface Features

Automatic Summarization. GraphWhisper generates detailed chart summaries immediately after image upload (Fig. 2A), identifying chart type, describing data ranges, and highlighting patterns. Summaries adapt to complexity.

Conversational Querying. Users enter questions via text or voice. Queries are classified into four categories [17]: analytical, visual, contextual, and navigation. Responses appear with confidence indicators (Fig. 2B).

Dynamic Follow-up Questions. The system generates contextual suggestions based on chart content (Fig. 2D), supporting progressive exploration.

Multimodal Input. Voice recognition supports seamless switching between text and voice modalities (Fig. 2E).

4.4 Confidence Communication

GraphWhisper provides explicit confidence assessment considering data extraction clarity, question complexity, and image quality. Confidence uses descriptive language (e.g., "GraphWhisper is 100 percent confident" or "70 percent confident") rather than numerical scores.

5 User Study

5.1 Method

We conducted remote Zoom sessions with 15 BLV participants recruited through the National Federation of the Blind. Nine were blind and six had low vision; eleven used JAWS, three used NVDA, and one used VoiceOver. Each 45-minute session included system introduction, exploration of multiple charts from our prepared pool, and post-session interviews.

5.2 Results

Usability and Satisfaction. All participants rated the system exceptionally high across all dimensions (4.87-5.0/5.0): ease of use (4.87), quality of descriptions (5.00), usefulness of Q&A (4.87), comparison to usual methods (4.87), and overall satisfaction (4.93). Minimal standard deviations (0.00-0.34) indicate strong consensus. All participants used every feature, demonstrating universal adoption.

Question-Answering Accuracy. Participants asked 185 questions total across diverse chart types and analytical tasks. GraphWhisper achieved 94% overall accuracy (174/185 correct), with consistent performance across query types: analytical 94% (73/78 questions), visual 94% (49/52 questions), contextual 94% (31/33 questions), and navigation 95% (21/22 questions). This distribution aligns

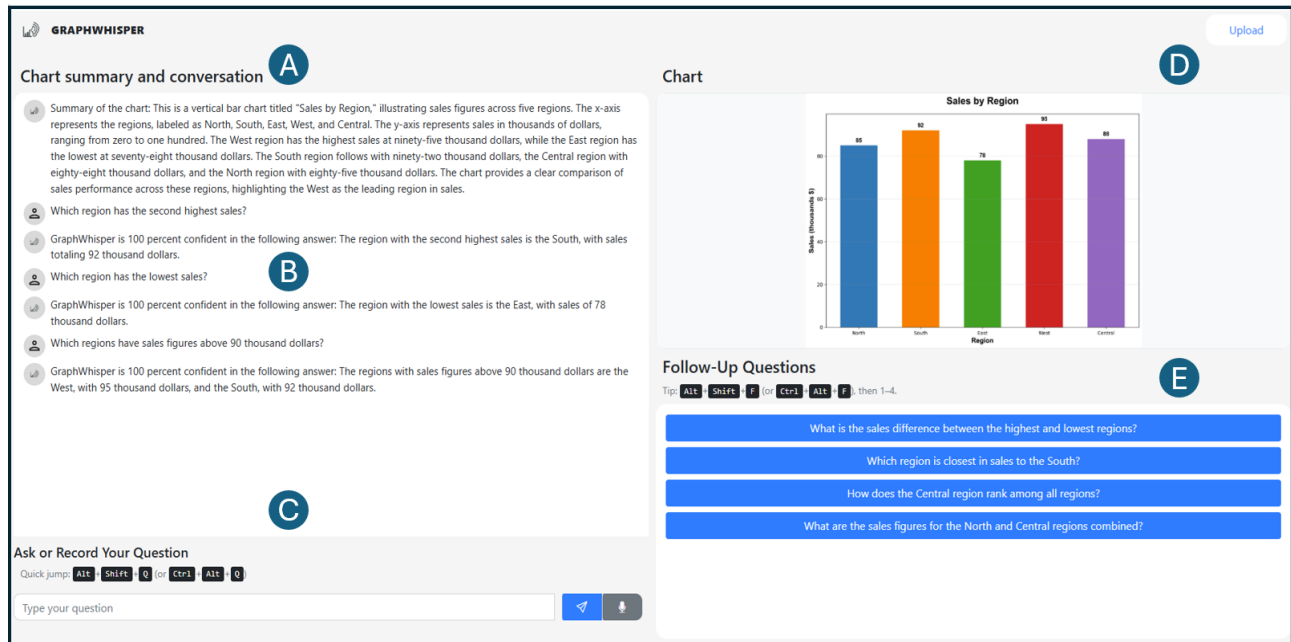


Figure 2: GraphWhisper user interface showing (A) automatic chart summary, (B) conversation area with confidence indicators, (C) chart visualization, (D) follow-up question suggestions, and (E) multimodal input.

with Kim et al.'s [17] findings, confirming authentic BLV interaction patterns. The consistency across query types demonstrates that our chart-type-specific methodology generalizes well across different question categories, supporting both data extraction tasks and visual interpretation queries.

Feature Adoption. All features received strong ratings: automatic summaries (4.9/5.0, 100% found helpful), follow-up questions (5.0/5.0, 100% helpful), and confidence indicators (4.9/5.0, 100% helpful). Participants valued independence: *"This lets me explore on my own and ask the specific questions I'm interested in."* Users appreciated response quality: *"The answers give me the detail I need for my work."* And accessibility: *"The system was intuitive and very easy to use."* The universal feature adoption (100% used every feature) demonstrates that all interface components addressed genuine user needs rather than adding unnecessary complexity.

6 Discussion

6.1 Key Findings

Our enhanced Charts-of-Thought methodology demonstrates that chart-type-specific analysis substantially improves LLM performance (98.1% VLAT accuracy), addressing RQ1. GraphWhisper successfully processes real-world image formats while maintaining high accuracy (94% across user questions), addressing RQ2 and demonstrating that practical accessibility tools can achieve accuracy comparable to structured-data systems. Interface features—automatic summaries, follow-up suggestions, confidence indicators—effectively support BLV chart exploration needs (RQ3), with exceptional user satisfaction scores (4.87–5.0/5.0) validating their practical effectiveness [17].

6.2 Implications

GraphWhisper demonstrates that AI-powered approaches can address longstanding barriers in data visualization accessibility [24]. Practical chart accessibility tools are achievable using current LLM capabilities with effective prompting [9, 32], without requiring specialized model training. Our findings support conversational interfaces as valuable complements to traditional approaches like keyboard navigation [2], alternative text [27], and sonification [35]. The enhanced methodology provides a framework applicable to other visual interpretation tasks [3, 13], while our integration of BLV user needs with systematic LLM evaluation provides a model for bridging user-centered design with technical development [12, 17].

6.3 Limitations

GraphWhisper works best with high-resolution charts but struggles with compressed images (accuracy declines to 68% for images <150 DPI). Complex multi-series visualizations and unusual encoding schemes present challenges [15, 26]. Our 15-participant study may limit generalizability. The system supports only static images and assumes visualization familiarity in English-speaking users.

7 Conclusion

We presented GraphWhisper, a conversational chart accessibility system enabling BLV users to interact with visualizations through natural language. By enhancing Charts-of-Thought with chart-type-specific procedures and processing standard image formats, GraphWhisper addresses critical limitations of existing tools requiring structured data, achieving strong technical performance (98.1% VLAT, 94% user study accuracy) and exceptional user satisfaction

(4.87-5.0/5.0). Future work will expand chart type coverage, enhance color interpretation, and integrate GraphWhisper into existing visualization workflows.

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